Vulnerability Assessment Report

Student Name: **Aditi Devendra Bansod**
Task: **Week 2 – VAPT Practical Task**
Tools Used: **Kali Linux, Nmap, Nikto**
Target Machine: **Metasploitable2**

Introduction

Vulnerability Assessment and Penetration Testing (VAPT) is a security testing process used to identify weaknesses in computer systems, networks, and applications. The objective of this task is to perform vulnerability scanning on a vulnerable machine (Metasploitable2) using security tools such as Nmap and Nikto. These tools help identify open ports, running services, and possible security misconfigurations.

Tools Used

Tool	Purpose
Nmap	Network scanning and service detection
Nikto	Web server vulnerability scanning
Kali Linux	Security testing environment
Metasploitable2	Vulnerable target machine

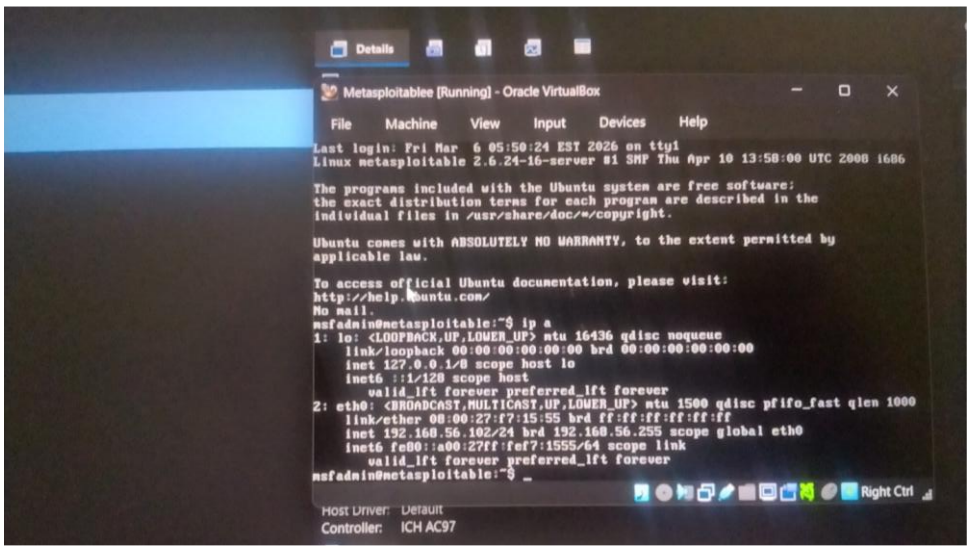

Figure 1: Identification of Target IP Address

Target Information

Target Machine: Metasploitable2

Parameter	Value
IP Address	192.168.56.102
Operating System	Linux
Purpose	Vulnerability Testing

- **Nmap Scan Results:-** Nmap was used to identify open ports and services running on the target machine.

  Command used: -nmap -sV 192.168.56.101

Port	Service	Version	Risk
21	FTP	vsftpd 2.3.4	High
22	SSH	OpenSSH	Medium
80	HTTP	Apache	Medium
445	SMB	Samba	High

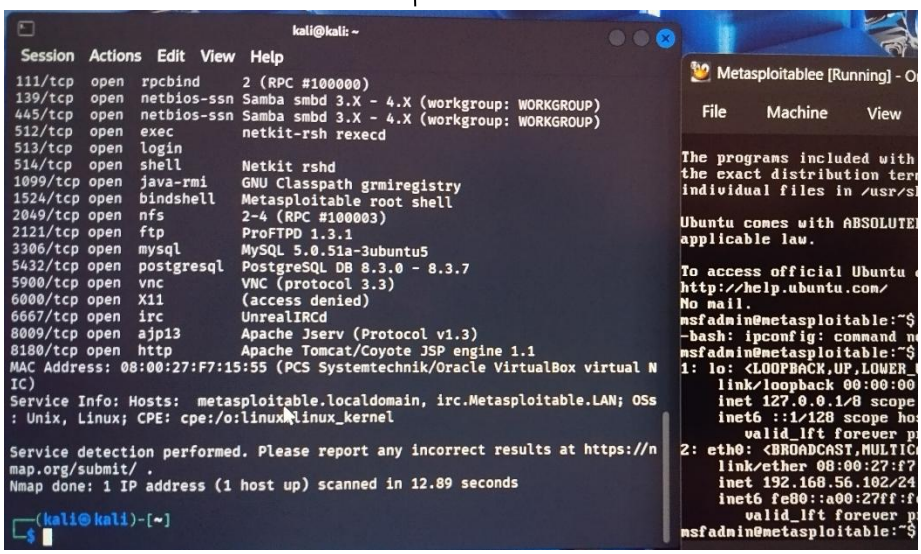

Figure 2: Nmap Network Scan Results.

- **Nikto Scan Results:-**Nikto was used to scan the web server for vulnerabilities and misconfigurations.

Command used: - nikto -h http://192.168.56.101

Vulnerability	Description	Risk
Outdated Apache	Old server version	Medium
Directory indexing enabled	Sensitive files accessible	Medium
Missing security headers	Weak web security configuration	Low

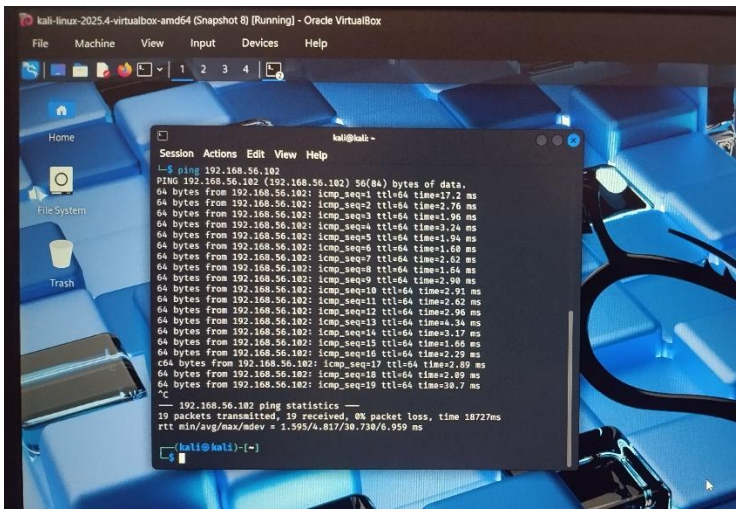

Figure 3: Nikto Web Server Vulnerability Scan

Vulnerability Prioritization

Scan ID	Vulnerability	Priority
001	FTP Anonymous Login	High
002	Open SMB Port	Medium
003	Outdated Apache	Medium

Remediation

Example:

- Update Apache to the latest secure version.
- Disable unnecessary open ports.
- Disable anonymous FTP access.
- Implement proper web security headers.

Conclusion

The vulnerability scanning process identified several security weaknesses in the target system including open ports, outdated services, and web server misconfigurations. These vulnerabilities could potentially be exploited by attackers. Proper patch management, service hardening, and network security controls should be implemented to reduce the risk.